

Currently we have multiple methods for cryptography which have different purposes. The primary method would be Elliptic Curves for public-key cryptography, ensuring that information sent over the communication channel is well protected. The start and end device will use Hashing like SHA-512 and Hashed Message Authentication Codes (HMACs) to ensure integrity verification of the data.

Fig. 1: Authentication Channel

Authentication goal:

- ▶ Bob wants to be sure that only Alice could have sent the message

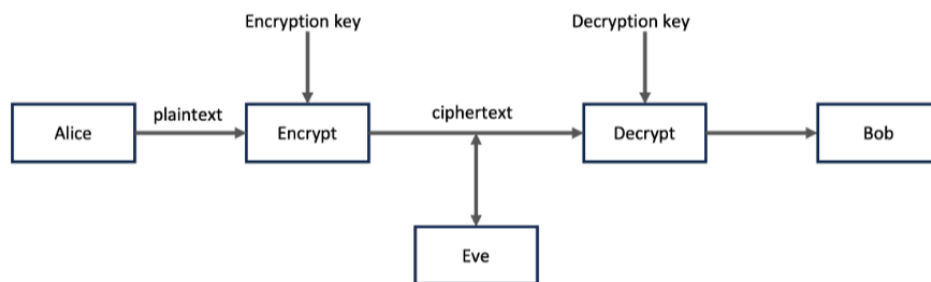
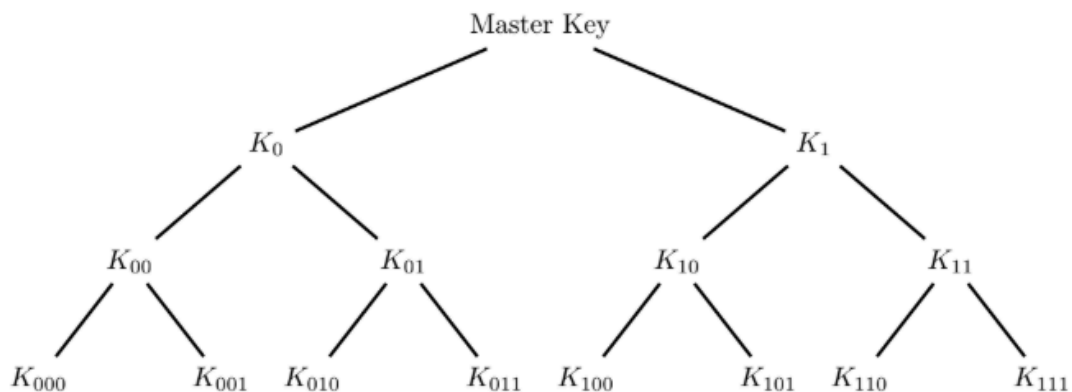


Fig. 2: Key Sharing Tree



Generated cryptographic secrets must be securely stored in nonvolatile storage to ensure data-at-rest protection. A PRF-based Key Derivation Tree (GGM Tree) will be used with global.secrets as the root node. Each subsequent child node is deterministically created through the extract-expand paradigm of HKDF. The group id is used by the HSM to derive the path of the Group Key. The user provides a 6-character hex PIN for each HSM called HSM_PIN. The verification value digest is calculated through HMAC(GroupKey, HSM_PIN) and it is stored in flash as a "golden reference" and recalculated from the tree whenever access is requested.

MPU:

A memory protection unit (MPU) can be utilized to enhance security, providing robust memory protection mechanisms. The MPU supports up to eight configurable memory regions, enabling granular access control. Additionally, the Sub Region Disable (SRD) feature allows for efficient use of memory regions by selectively disabling specific sub-regions. Furthermore, the MPU can enable a background region that enforces default memory map attributes, ensuring a secure and consistent memory access policy. Memory security can also be tested at compile time through tools like Valgrind to minimize the risk of memory-related vulnerabilities.

Crypto Hash:

The system leverages WolfSSL/Simple Crypto, which provides a suite of cryptographic algorithms available for implementation. In the absence of a Trusted Third Party (TTP) (Fig. 1), a PRF-based Key Derivation Tree rooted in global.secrets is used. Authentication is facilitated through an HMAC-SHA256 protocol that requires a cryptographic group key per HSM accompanied by a user PIN. For secure key exchange, the Elliptic Curve key exchange protocol is employed to establish shared secrets between communicating parties.

How CryptoHash is implemented on SoC:

Cryptographic operations are integrated into the SoC via a hardware-assisted derivation and verification sequence. SHA-256 is utilized for GGM tree expansion and Group ID verification. Access attempts involving invalid PINs or unauthorized Group IDs trigger a mandatory 5-second hardware timer delay to mitigate brute-force attacks. Data-at-rest protection is achieved by storing a "Golden Reference" HMAC digests in non-volatile storage, ensuring plaintext HSM_PIN secrets are never exposed in flash memory.